

MOSJOEN KJAERSTAD, Norway

WMO: 011220

Lat: 65.784N

Lon: 13.218E

Elev: 237

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	-0.7	-11.1	3.0	-4.8	-6.1	4.0	1.1	23.2	31.3	20.5	31.2	3.9	170	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	77.4	61.5	73.0	59.6	69.5	57.7	63.7	73.6	61.3	70.2	59.1	67.0	7.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.5	76.5	67.8	57.3	70.7	64.9	55.3	65.7	62.6	29.1	73.6	27.4	70.4	25.9	67.3	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	15.5	13.1	DB	-7.8	83.4	5.4	5.1	-11.7	87.0	-14.8	90.0	-17.9	92.9	-21.8	96.6	
			WB	-8.2	65.8	5.0	3.0	-11.8	68.0	-14.7	69.7	-17.5	71.4	-21.2	73.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	22.3	23.9	29.0	37.2	45.7	51.7	58.1	55.4	49.7	39.2	29.7	25.7	
	DBStd	14.83	11.43	11.39	8.55	5.29	6.92	6.16	6.32	4.78	5.42	6.79	10.07	11.27	
	HDD50	4627	857	730	652	385	176	48	2	7	70	338	608	754	
	HDD65	9493	1322	1150	1117	835	600	401	233	299	460	799	1058	1219	
	CDD50	634	0	0	0	0	41	100	254	175	60	4	0	0	
	CDD65	25	0	0	0	0	0	3	20	2	0	0	0	0	
	CDH74	322	0	0	0	0	22	31	237	31	1	0	0	0	
	CDH80	74	0	0	0	0	2	6	65	1	0	0	0	0	
Wind	WSAvg	5.2	5.0	5.3	6.0	5.4	5.7	5.7	5.2	4.6	4.4	4.8	4.7	5.1	
Precipitation	PrecAvg	66.3	7.2	6.8	6.8	4.3	3.1	3.3	3.4	3.7	6.5	7.0	6.2	7.9	
	PrecMax	91.2	15.5	14.8	12.8	10.0	5.0	7.6	8.4	7.5	18.5	14.3	12.7	14.8	
	PrecMin	46.9	0.6	0.9	1.3	1.2	1.0	0.7	1.4	0.8	1.7	2.4	0.6	1.4	
	PrecStd	10.3	3.9	3.9	3.5	2.4	0.9	1.7	1.7	1.4	3.5	3.1	3.5	4.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.2	45.6	46.6	63.0	77.4	78.3	87.4	77.8	70.2	59.3	50.6	46.8	
		MCWB	43.5	41.8	40.5	48.0	56.4	60.3	66.9	61.5	55.5	50.9	47.9	44.2	
	2%	DB	41.3	43.2	44.4	53.8	70.5	71.8	80.5	73.3	65.0	55.5	47.9	44.4	
		MCWB	38.3	40.3	40.2	43.2	54.5	57.0	63.9	60.1	54.0	47.7	45.1	41.7	
	5%	DB	38.7	41.2	42.2	49.1	65.8	68.0	75.3	69.5	62.0	52.2	43.6	41.1	
		MCWB	36.2	38.3	38.2	41.2	52.7	55.4	61.8	58.5	52.7	47.2	40.7	38.0	
	10%	DB	35.8	38.5	40.3	46.0	59.0	64.3	70.8	65.8	58.7	49.1	40.7	38.6	
		MCWB	34.1	35.6	36.6	39.8	49.5	53.8	59.5	57.0	52.4	45.4	38.3	36.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.9	42.3	42.8	48.5	58.3	62.5	68.4	64.5	59.0	51.6	48.8	44.5
			MCDB	45.6	44.6	45.4	61.6	73.3	76.0	81.0	72.2	65.4	57.5	50.5	46.4
2%		WB	38.8	40.7	40.5	44.9	55.6	59.4	65.4	61.7	56.6	49.5	44.8	42.0	
		MCDB	40.7	43.1	43.1	51.6	68.0	68.9	77.1	70.1	62.2	53.7	47.6	44.0	
5%		WB	36.1	38.7	38.7	42.4	53.0	57.0	63.1	59.7	54.7	47.5	41.1	38.2	
		MCDB	37.9	41.0	41.6	46.8	64.1	65.3	73.1	67.4	59.6	51.4	43.0	40.1	
10%		WB	34.1	35.9	36.8	40.6	49.9	54.9	60.7	57.9	52.9	45.8	38.7	36.2	
		MCDB	35.6	38.0	39.9	44.8	58.8	62.4	69.1	64.7	57.3	49.0	40.3	38.2	
Mean Daily Temperature Range	5% DB	MDBR	10.4	11.5	13.9	14.4	16.4	14.2	15.1	16.3	14.8	11.0	8.7	10.0	
		MCDBR	9.6	10.7	11.9	20.8	27.2	23.1	24.5	24.5	20.9	13.3	10.4	10.4	
		MCWBR	8.2	8.9	8.6	12.1	13.2	11.2	10.5	12.5	11.3	9.1	8.5	8.5	
	5% WB	MCDBR	9.2	10.9	10.5	16.7	25.2	20.5	21.8	21.4	16.8	11.7	10.5	9.9	
		MCWBR	8.1	9.5	8.3	11.0	12.8	10.4	10.2	11.8	10.5	8.4	8.8	8.4	
Clear-Sky Solar Irradiance	taub	N/A	0.278	0.311	0.338	0.343	0.343	0.352	0.338	0.321	0.295	0.353	N/A		
	taud	N/A	2.142	2.196	2.243	2.363	2.473	2.475	2.505	2.535	2.353	2.615	N/A		
	Ebn at Noon	N/A	184	237	261	272	276	269	262	241	183	76	N/A		
	Edh at Noon	N/A	17	27	33	32	30	29	26	20	14	6	N/A		
All-Sky Solar Radiation	RadAvg	24	143	545	1071	1420	1529	1414	1099	583	232	47	5		
	RadStd	3	28	86	116	103	192	167	121	82	43	7	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	+0.63	N/A	N/A	N/A	N/A	N/A	-167	-239	N/A	N/A

Nomenclature: See separate page